Rashmiprakash Satapathy

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Professional Summary

Software Engineer with experience in building full-stack and backend systems using **React.js**, **Django**, **Node.js**, **Python**, and SQL/NoSQL databases. Skilled in **REST API** design, distributed and concurrent systems, database optimization, and **CI/CD**, with a strong foundation in **C++**, **Data Structures & Algorithms**, and **Object-Oriented Programming**.

Experience

Mercados Energy Markets India Limited | *Analyst* | *Software Engineer* | *Full-Stack* July 2025 – Present

- Built **Django REST APIs** and **React.js/Redux** features for UPPCL's production bill verification platform, supporting billing workflows across **300+ power plants** and **1M+ billing records** yearly.
- Engineered **plant-specific billing and validation logic** in Django to handle varied data formats and calculation rules, reducing bill processing time and manual effort by **50%** across a **4-stage billing workflow**.
- Optimized **MongoDB queries, aggregation pipelines, and bulk-write operations** over a **100K+ record dataset**, reducing database query latency by **20%** and improving processing throughput.
- Refactored backend business logic into **18 modular service-layer components**, eliminating duplicated code and simplifying plant-specific extensions.
- Implemented **API validation, error handling, and automated tests** across billing workflows, preventing failed requests and billing discrepancies.
- Built a **Python resource adequacy optimization tool using PuLP** (linear programming) to solve multi-constraint energy planning problems, cutting scenario computation time by **30%** across **15+ scenarios**.

Mercados Energy Markets India Limited | *Internship* | *Machine Learning* | *Python* Dec 2024 – June 2025

- Developed a **Python ETL pipeline** processing **10M+ SCADA, weather, and electricity market records**, reducing data preprocessing time by **25%** through automated data preparation and integration.
- Engineered temporal and weather features and trained **XGBoost, Random Forest, and stacking ensemble models** for day-ahead electricity demand forecasting, achieving **under 3% MAPE**, **12% below baseline**.
- Automated demand-monitoring reports using **Python and Power BI**, cutting manual reporting effort by **40%**.
- Implemented **data cleaning, validation, and transformation workflows** across **100K+ monthly records**, eliminating invalid and missing entries before model training.

Technical Skills

Languages: C++, Python, Java, SQL, JavaScript, Typescript
Core CS: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), System Design, DBMS
Backend & Frontend: Django, FastAPI, Node.js, Express.js, REST APIs, React.js, Redux, Tailwind CSS
Testing & Practices: Unit Testing, Integration Testing, pytest, Git, GitHub, Agile
Databases & Infrastructure: PostgreSQL, MySQL, MongoDB, Redis, Data Modeling, AWS, Docker, CI/CD
AI/ML: XGBoost, Scikit-learn, LangGraph, Vector Embeddings, Semantic Search, RAG, LLM Integration

Projects

Asynchronous Task Queue Engine | *Node.js, Express, PostgreSQL, Redis, Docker* [GitHub]

- Architected a horizontally-scalable, concurrent task queue in **Node.js**, **PostgreSQL**, and **Redis** with **priority-based job queues** and row-level locking for safe parallel job consumption across multiple workers.
- Engineered fault tolerance with a **dead-letter queue**, automated retries, a **Redis-backed circuit breaker**, and stale-job recovery; **containerized** all services with **Docker Compose** and exposed **Prometheus**-format metrics.

Multi-Agent Chat System | *React, FastAPI, MongoDB, LangGraph* [GitHub] [Live Site]

- Built a full-stack chat platform with a **supervisor-agent architecture** routing queries across **3 specialized sub-agents** (RAG, NL-to-SQL, Web Search), **real-time SSE token streaming**, and a Pinecone vector search.
- Secured the system with **JWT authentication** and persistent multi-turn memory via **PostgreSQL-backed checkpointers**, with end-to-end distributed tracing across agent/tool calls using LangSmith.

Education

National Institute of Technology Raipur 2021-2025
Bachelor of Technology (CGPA: 8.36) *Raipur, Chhattisgarh*

Achievements & Coding Profiles

- **LeetCode**: **500+** questions solved | Contest Rating: **1927 (Knight)**; **GeeksforGeeks**: **400+** questions solved.
- Achieved **97.53 percentile** in JEE Main 2021 and **94.4%** in CBSE Class XII Board Examinations.
- Secured **1st Rank** in Electrical Engineering at **Vigyaan**, a National Level Science Exhibition organized by NIT Raipur.